

Child Malnutrition in Africa, with a Focus on Ghana

From the continent, to Ghana, to its 16 regions

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Introduction

Malnutrition remains one of the defining health challenges in Africa, but the picture is far from uniform. This report zooms in across three levels: where **Ghana sits among African countries**, how malnutrition has **changed within Ghana**, and how it is **distributed across Ghana's 16 regions** and social groups. Ghana is an instructive case: a relative success story on chronic undernutrition that still carries sharp internal inequalities.

The headline indicator is **stunting** (low height-for-age, the standard measure of chronic child undernutrition), complemented by underweight, wasting, overweight and anaemia.

Data and methods

All data come from open sources, with no API keys required:

- **DHS Program API** - Ghana stunting by region and by wealth, education and residence, across survey rounds (1993–2022).
- **World Bank Open Data API** - malnutrition indicators for African countries and Ghana's national trend.
- **geoBoundaries** - Ghana's administrative (region) boundaries for mapping.

Analysis is descriptive and equity-focused: continental benchmarking, national trends, a regional choropleth, and disparity ratios across social groups.

```
afr <- read_csv("data/africa_stunting_latest.csv", show_col_types = FALSE)
trend <- read_csv("data/ghana_trend.csv", show_col_types = FALSE)
region<- read_csv("data/ghana_stunting_region.csv", show_col_types = FALSE)
equity<- read_csv("data/ghana_equity.csv", show_col_types = FALSE)
gh_sf <- st_read("data/ghana_adm1.geojson", quiet = TRUE) |>
  mutate(region = str_remove(shapeName, " Region$"))
```

Africa context: where does Ghana stand?

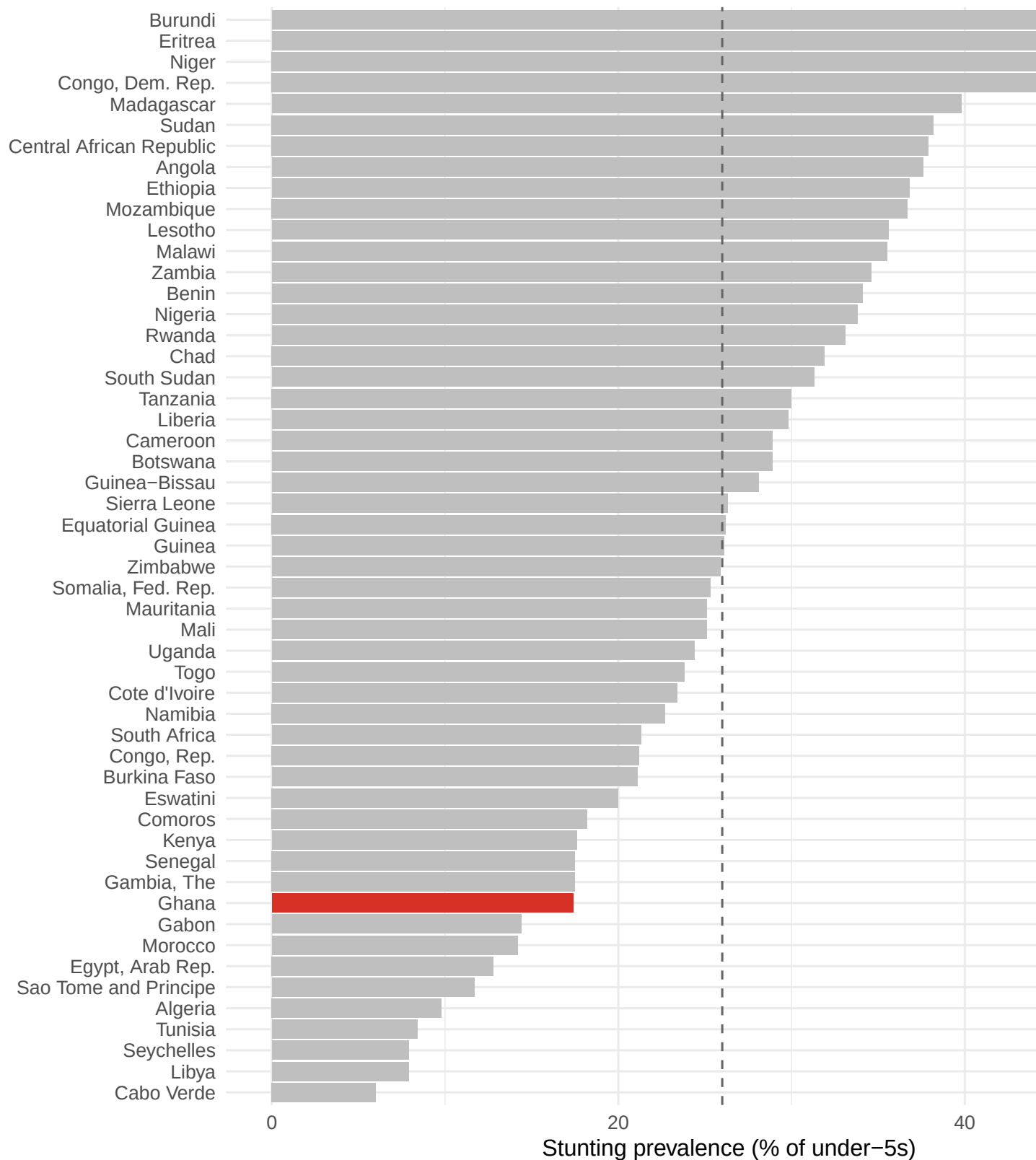
```

afr <- afr |> mutate(is_ghana = iso3 == "GHA")
ghana_val <- afr$value[afr$iso3 == "GHA"]
afr_median <- median(afr$value)
ggplot(afr, aes(value, fct_reorder(country, value), fill = is_ghana)) +
  geom_col() +
  geom_vline(xintercept = afr_median, linetype = "dashed", colour = "grey40") +
  scale_fill_manual(values = c(`FALSE` = "grey75", `TRUE` = "red"), guide = "none") +
  labs(title = "Child stunting across African countries (latest available)",
        subtitle = sprintf("Ghana = %.0f%% (red); African median = %.0f%% (dashed). Lower is better.",
                             ghana_val, afr_median),
        x = "Stunting prevalence (% of under-5s)", y = NULL)

```

Child stunting across African countries (latest available)

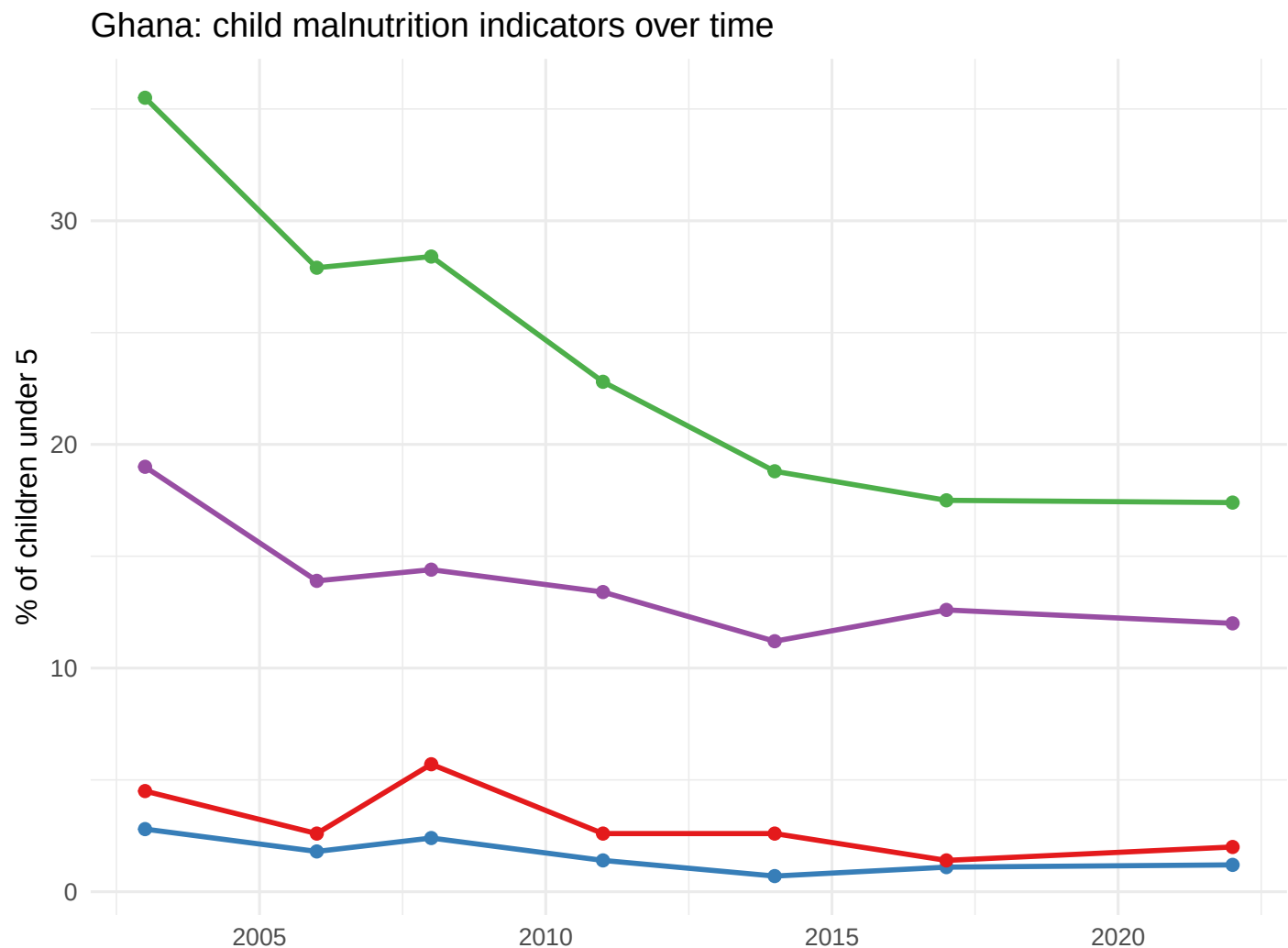
Ghana = 17% (red); African median = 26% (dashed). Lower is better.



Ghana's stunting prevalence (17%) is **well below the African median** of 26%, placing it among the continent's stronger performers on chronic undernutrition.

Ghana over time: progress and the double burden

```
trend |>
  filter(indicator %in% c("stunting", "underweight", "overweight", "severe_wasting")) |>
  ggplot(aes(year, value, colour = indicator)) +
  geom_line(linewidth = 1) + geom_point(size = 2) +
  scale_colour_brewer(palette = "Set1") +
  labs(title = "Ghana: child malnutrition indicators over time",
       x = NULL, y = "% of children under 5", colour = NULL)
```



Chronic undernutrition (stunting, underweight) has fallen substantially, yet **wasting has not disappeared and overweight persists** - the early signs of a “double burden” of malnutrition that increasingly characterises middle-income transitions.

Within Ghana: the regional divide

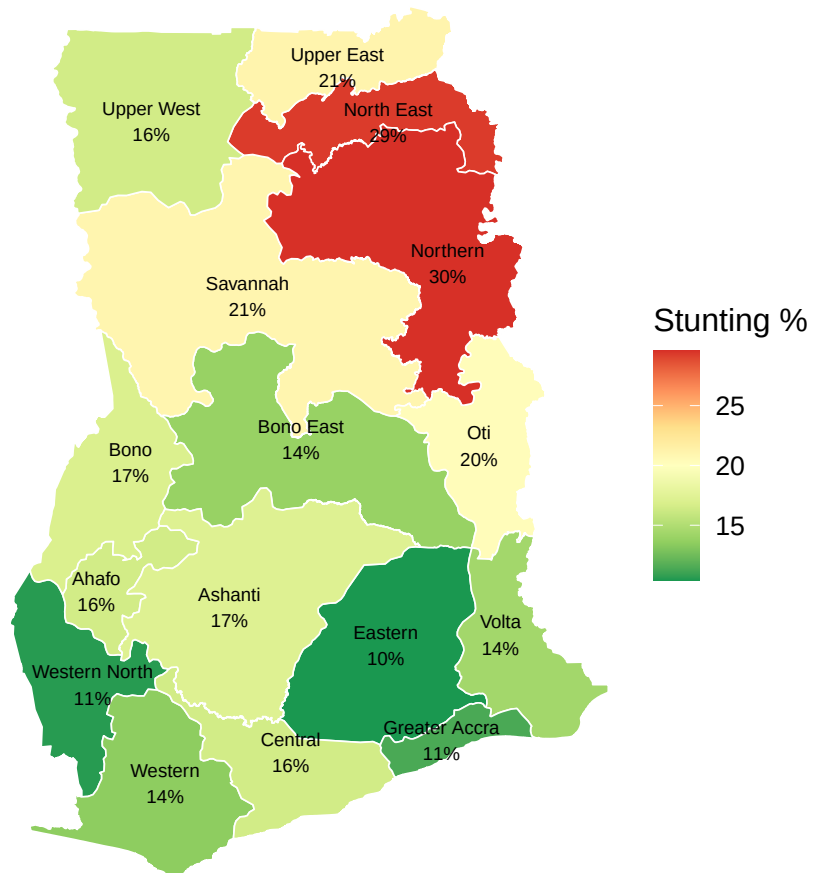
We map stunting across Ghana's 16 regions using the most recent survey (2022). DHS reports a mix of old and new regions; we select the 16 current regions and match them to the boundary file.

```
# Map DHS 2022 leaf labels -> boundary region names
m <- c(..Western (post 2022)="Western", ..Western North="Western North",
      "Central"="Central", "Greater Accra"="Greater Accra",
      "..Volta (post 2022)"="Volta", "..Oti"="Oti", "Eastern"="Eastern",
      "Ashanti"="Ashanti", "..Ahafo"="Ahafo", "..Bono"="Bono",
      "..Bono East"="Bono East", ....Northern(post 2022)="Northern",
      ....Savannah="Savannah", ....Northeast="North East",
      "..Upper East"="Upper East", "..Upper West"="Upper West")
reg22 <- region |> filter(survey_year == 2022, label %in% names(m)) |>
  mutate(region = m[label]) |> select(region, stunting = value)

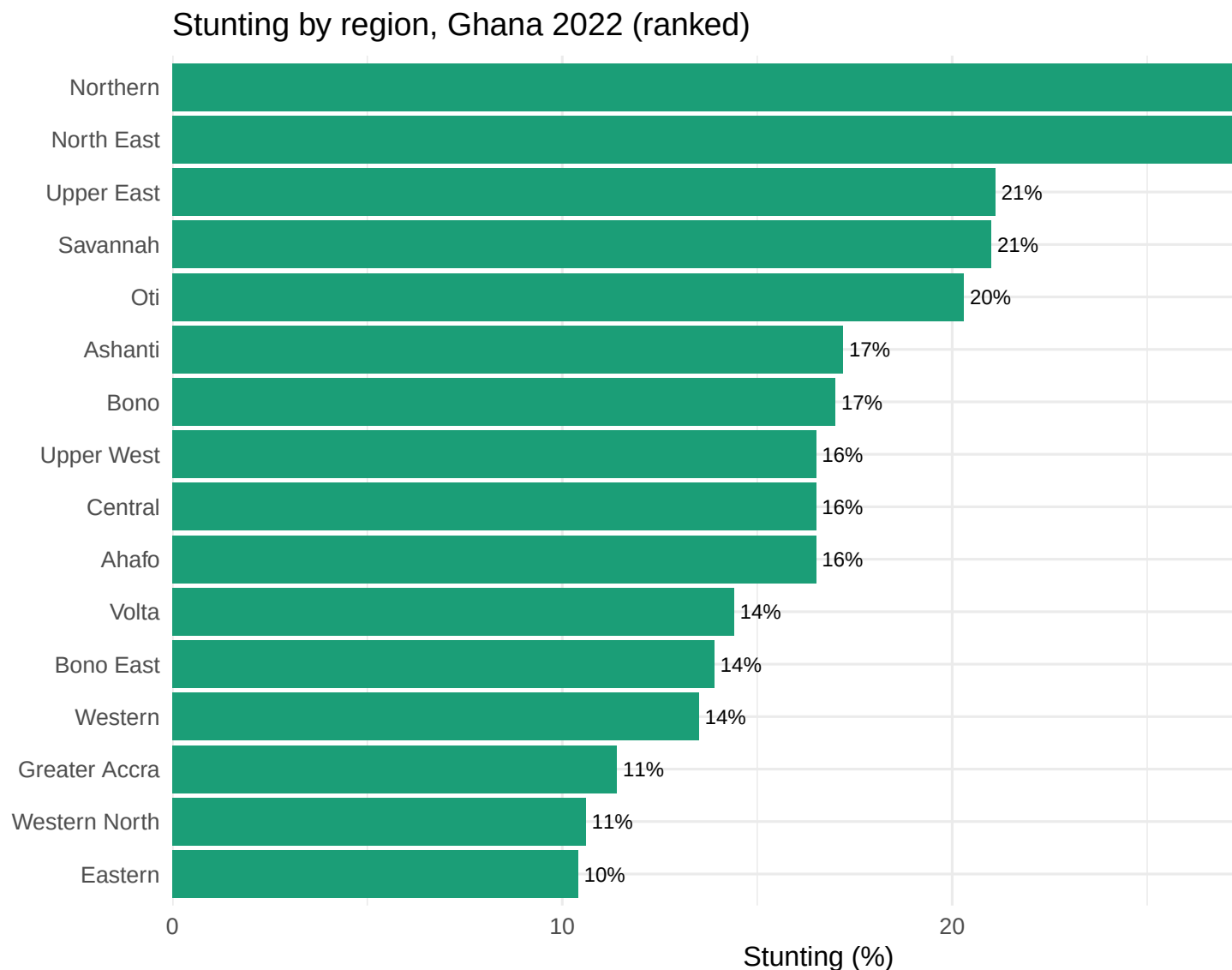
map_df <- gh_sf |> left_join(reg22, by = "region")
ggplot(map_df) +
  geom_sf(aes(fill = stunting), colour = "white", linewidth = 0.3) +
  scale_fill_distiller(palette = "RdYlGn", direction = -1, name = "Stunting %") +
  geom_sf_text(aes(label = paste0(region, "\n", round(stunting), "%")), size = 2.4) +
  labs(title = "Child stunting by region, Ghana 2022",
       subtitle = "The north carries a far heavier burden than the south") +
  theme_void(base_size = 12)
```

Child stunting by region, Ghana 2022

The north carries a far heavier burden than the south



```
reg22 |>
  ggplot(aes(stunting, fct_reorder(region, stunting))) +
  geom_col(fill = ACC) +
  geom_text(aes(label = paste0(round(stunting), "%")), hjust = -0.15, size = 3) +
  scale_x_continuous(expand = expansion(mult = c(0, 0.12))) +
  labs(title = "Stunting by region, Ghana 2022 (ranked)",
       x = "Stunting (%)", y = NULL)
```

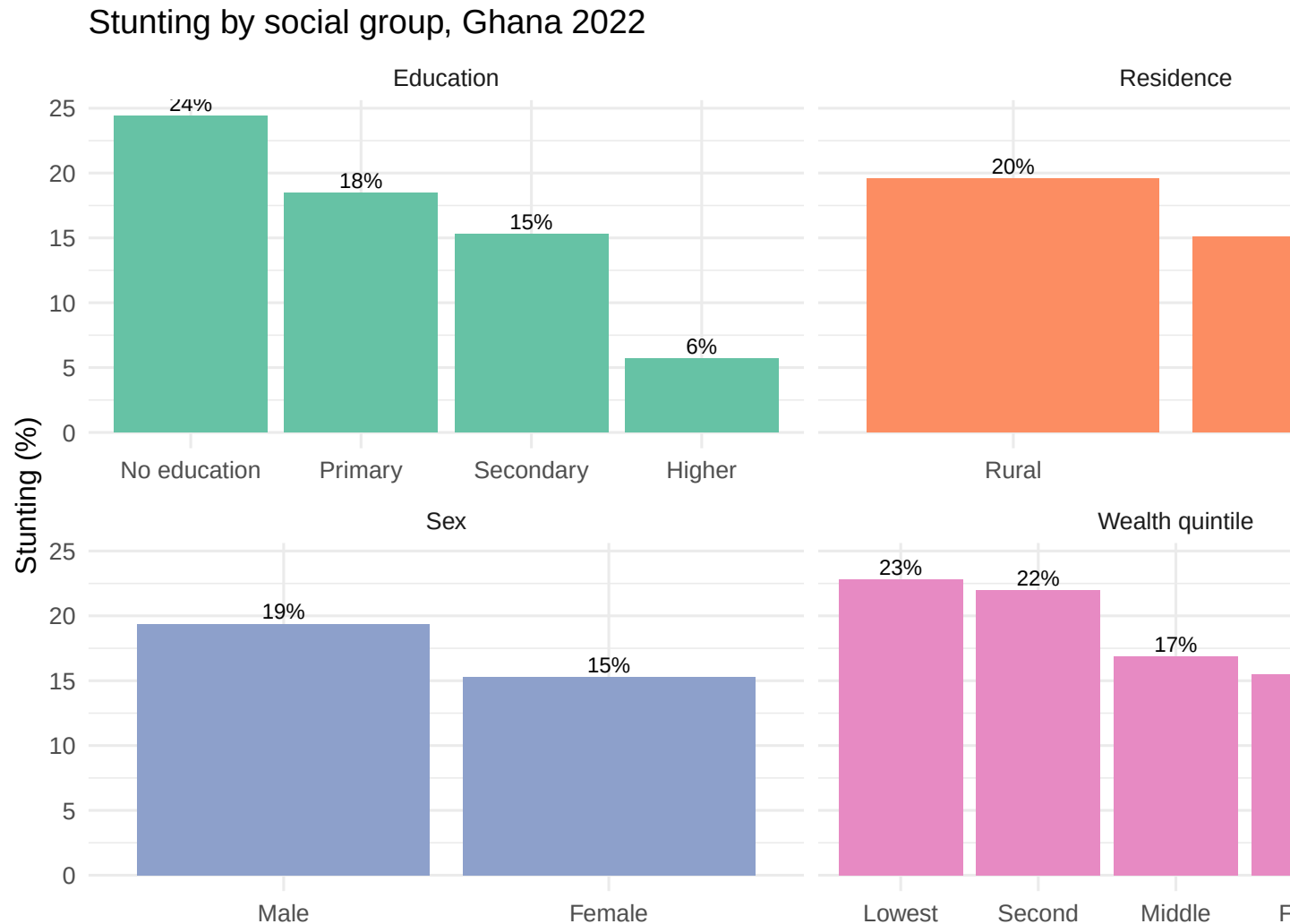


The gap is stark: stunting ranges from about **10% in Eastern** to **30% in Northern** - a 2.8-fold difference between the best and worst regions, reflecting Ghana's well-documented north-south divide.

Inequality by wealth, education and residence

```
eq <- equity |>
  filter(survey_year == 2022, indicator == "stunting",
         category %in% c("Wealth quintile", "Education", "Residence", "Sex")) |>
  mutate(label = factor(label, levels = c(
    "Lowest", "Second", "Middle", "Fourth", "Highest",
    "No education", "Primary", "Secondary", "Higher",
    "Rural", "Urban", "Male", "Female")))
ggplot(eq, aes(label, value, fill = category)) +
  geom_col(show.legend = FALSE) +
```

```
geom_text(aes(label = paste0(round(value), "%")), vjust = -0.3, size = 3) +
facet_wrap(~ category, scales = "free_x") +
scale_fill_brewer(palette = "Set2") +
labs(title = "Stunting by social group, Ghana 2022",
      x = NULL, y = "Stunting (%)")
```



The social gradient is steep. Children in the **poorest** wealth quintile are about **3 times** as likely to be stunted as those in the **richest** (23% vs 8%), and children of mothers with **no education** are around **4.3 times** as likely as those whose mothers have higher education. Rural children fare worse than urban ones. These gaps show that national averages hide who is being left behind.

Conclusion and limitations

Conclusion. Ghana is a relative success on chronic child undernutrition, outperforming most African countries and reducing stunting over two decades. But the national average conceals a sharp **north–south divide** and steep **wealth and education gradients**. Progress now depends less on national growth than

on **reaching the poorest regions and households** - and emerging overweight signals a coming double burden.

Limitations. Survey estimates carry sampling error and are collected only every few years; region definitions changed in 2019, complicating long-run regional comparisons; and this is an ecological, descriptive analysis, so associations should not be read as causal.

Built in R (tidyverse, sf, ggplot2) from DHS, World Bank and geoBoundaries open data. Author: Kingsley Amegah.